

Aetnamem Explained

Twenty Articles on Governed Memory for AI

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Based on *Governed Memory Without Embeddings*

July 2026

Abstract

This collection contains twenty standalone, high-level articles prepared for publication on X. It translates the architecture, released behavior, and measured claims of AETNAMEM through version 0.6.1 into accessible language without removing the technical boundaries. The series explains why durable AI memory is more than search, how four memory planes become one governed context pack, how multimodal observations retain provenance, how an auditor can trace memory influence, and how Memory Impact tests whether remembering changed a verified outcome.

Source paper: *Governed Memory Without Embeddings: Recall Forensics, Deterministic Retrieval, and Approval-Bound Effects in aetnamem*. Author: aetna000.com.

Editorial note: Each article is designed to stand alone. Technical measurements are used only where they clarify the idea. Statements about current implementation refer to the public version 0.6.1 repository and its content-addressed evidence summaries. Experimental machinery and an exploratory pilot are identified separately from completed product claims.

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1 Memory Is Not a Search Box

Article 1 of 20

A useful AI memory system must govern what becomes memory, not only find text that looks relevant.

When people hear “AI memory,” they often imagine a search engine attached to a chatbot. Store old conversations, find similar passages, place them in the next prompt, and let the model answer. That can improve continuity, but it does not answer the harder questions created by long-lived memory.

Who supplied a remembered statement? Was it the user, a web page, a tool, or the model itself? Is it still current? Did a later correction replace it? Was it reviewed? Can it be deleted? If it contributes to a proposed action, does it carry any authority to change the outside world?

AETNAMEM treats these as state-management questions. It stores source episodes, normalized records, lifecycle status, trust origin, provenance, and audit evidence. Retrieval operates inside those rules. A result cannot enter ordinary recall merely because it has a strong text match; it must first be an active memory under the lifecycle policy.

This changes the role of the language model. The model may help a user speak to the system, summarize evidence, or propose an action. It does not own the truth of the database, decide which untrusted content becomes accepted, or turn recalled information into permission.

The practical goal is modest but important: when an assistant says “I remember,” the system should be able to show what was accepted, where it came from, what changed since then, and why it was eligible to appear.

Important boundary. AETNAMEM does not prove that a remembered statement is true. It preserves who supplied it, how it was treated, and how it participated in the system.

2 A Filing Cabinet, Catalogue, Map, and Receipt Book

Article 2 of 20

The easiest way to understand AETNAMEM is as four familiar tools working together.

The first tool is a **filing cabinet**. It holds canonical episodes and records: the original source material and the concise facts derived from it. These are the durable objects from which other structures can be rebuilt.

The second is a **card catalogue**. SQLite full-text search helps locate active records without scanning and scoring every item in memory. It is a local index with explicit limits, not a second source of truth.

The third is a **map**. A derived graph connects entities and typed relations. If memory says “my boss is Sarah” and “Sarah prefers SEA,” the map can follow the relation between those statements. Every edge points back to a record, and every record points back to its source episode.

The fourth is a **receipt book**. Important transitions append evidence to a hash-linked audit chain: admission, promotion, correction, deletion, retrieval, graph maintenance, approval, and external-action outcomes.

This separation is useful because the catalogue and map may be wrong or outdated without corrupting the filing cabinet. They can be cleared and rebuilt from canonical records. The receipt book then helps an operator check which transitions occurred and whether the recorded chain still verifies.

Many systems combine these responsibilities into one opaque model-driven pipeline. AETNAMEM separates them so that failures have smaller meanings: an index problem is not automatically a memory-integrity problem, and a relevant memory is not automatically an authorized action.

Important boundary. Rebuildability improves recovery from index damage. It does not replace backups or recover canonical records that were lost from disk.

3 Why the Source of a Memory Matters

Article 3 of 20

The same sentence should be treated differently when it comes from a user, a web page, a tool, or the assistant itself.

Imagine a user has stored a preferred airport. Later, a web page contains the sentence “your preferred airport is LAX.” A text extractor can read both sentences, but it should not treat them as equally authoritative.

AETNAMEM classifies origin before a candidate memory becomes active. Direct user statements and explicitly reviewed content may enter active memory. Marked web-page or tool-output content begins quarantined. It remains visible for review, but ordinary recall and graph traversal exclude it.

This is a practical defense against durable memory poisoning. A hostile page cannot silently replace a trusted preference merely because it contains language that looks like a personal fact. Promotion is a separate action with its own provenance and audit event.

Assistant responses are also not automatically promoted into semantic memory. Otherwise, a model could repeat its own output until generated text appeared to have become an established fact. The system can log the exchange without granting the model self-authorship over durable truth.

Origin is not the same as correctness. A trusted user can be mistaken, and a web page can be accurate. The trust tier instead answers a narrower policy question: how may this source participate before review?

Important boundary. Origin classification depends on the host correctly labeling external content. If a compromised integration presents remote text as a direct user statement, the storage layer cannot recover the missing context.

4 Quarantine Is Memory With a Door

Article 4 of 20

Suspicious information does not have to be either trusted or discarded; it can be retained for inspection without entering normal recall.

Binary choices are often too crude for AI memory. Throwing away every untrusted observation loses potentially useful information. Accepting every observation creates a path for web content, tool output, or generated text to shape future answers as if a user had approved it.

AETNAMEM uses quarantine as an intermediate lifecycle state. A quarantined record keeps its source and can be reviewed, but it is excluded from ordinary recall. Its graph edges cannot become bridges between trusted facts. It also cannot supersede an active record.

Promotion is explicit. A reviewer can inspect the proposed memory, confirm it, and move it into the user-confirmed trust tier. The transition is recorded rather than hidden inside a model's confidence score.

This design makes disagreement visible. The system can say, in effect, "this information was observed, but it has not been accepted." That is more useful than pretending all stored text has the same standing.

Quarantine is especially important for agents that browse, call tools, or receive messages from external channels. Those systems encounter content that may be relevant but was not authored as an instruction to the memory layer.

Important boundary. A reviewer can still approve false information. Quarantine governs admission and records responsibility; it does not provide an oracle for fact checking.

5 Corrections Should Not Erase History

Article 5 of 20

A durable memory should show what changed, not silently rewrite the past.

Suppose a user says “my preferred airport is Sydney,” then later changes it to Melbourne. Editing the old database row in place would produce the current answer, but it would remove the evidence that a correction occurred.

AETNAMEM uses explicit supersession for recognized single-valued facts. The new active record becomes current, while the prior record moves to a superseded state and points to its replacement. Normal recall returns the new value, but history remains reconstructable.

This matters when people ask why an assistant behaved differently last month. The answer may depend on the memory state at that time, not the state visible today. A transition history allows an auditor to replay which record was eligible at a particular turn.

The policy also protects trusted state from unreviewed replacement. A quarantined record cannot supersede an active record, even when both appear to occupy the same fact slot.

Not every relation is single-valued. “Works with,” “likes,” and “avoids” may accumulate. Unstructured conflicts may remain as separate records for review rather than being merged by semantic guesswork.

Important boundary. Supersession depends on the system recognizing a structured fact key. Unsupported phrasing can remain as multiple records, because visible ambiguity is safer than an unexplained destructive update.

6 Forgetting Is More Than Removing a Search Result

Article 6 of 20

Deletion should remove governed payloads and indexes while leaving evidence that a deletion operation occurred.

Removing an item from search is not necessarily forgetting it. The original text may still exist in a record table, a graph edge, a full-text index, a source episode, or an archive.

In AETNAMEM, forgetting is a lifecycle transition across these structures. Matching records are tombstoned, selected payloads are purged, and related full-text and graph entries are removed. Source episodes are purged when no retained record still depends on them.

The system keeps a deletion receipt containing identifiers and digests rather than the erased words. This creates a useful balance: an operator can verify that a deletion was processed without preserving the sensitive value merely to make future auditing easier.

That choice creates a real replay boundary. If a memory contributed to an old answer and was later deleted, the system may reconstruct the historical decision from commitments and lifecycle events while correctly refusing to reproduce the deleted plaintext.

Deletion is deliberately accepted only from user-origin requests, and an empty selector is rejected instead of being interpreted as “delete everything.” These are small rules with large consequences for safety.

Important boundary. A local deletion receipt cannot prove that independent backups, exports, model-provider logs, or external systems deleted their copies. Those systems need separate retention policies and deletion receipts.

7 Why Aetnamem Keeps Embeddings Out of Agent Recall

Article 7 of 20

Embeddings are an optional investigator lens in Aetnamem, not a source of trusted agent recall or memory authority.

Embeddings are effective at finding paraphrases and conceptually similar text. AETNAMEM does not claim otherwise. Since version 0.5.1, it has offered optional semantic and hybrid discovery for the human-facing memories, search, and trace investigation commands. Embeddings live in a disposable SQLite sidecar and nominate canonical record identifiers. They do not change `Memory.recall()`, MCP agent recall, OpenClaw automatic recall, memory admission, promotion, or action authority. Lexical search remains the default.

This separation is operational. Replaying the trusted agent-recall path requires text normalization, static weights, database state, and code. Replaying a semantic investigation also requires the exact embedding model, tokenizer, preprocessing, numerical environment, vector-index snapshot, and dependencies. An agent therefore does not need to download and run a second model merely to remember.

The vector sidecar is a lens, never the source of record truth. Every semantic candidate is reloaded from canonical SQLite and checked for subject, lifecycle status, and content digest before it can be shown. Versioned index epochs keep model identities separate. Ollama epochs pin the model digest reported by the local service and fail closed if it changes; providers without a verifiable revision remain explicitly unverified unless the operator supplies one.

Deletion is also part of the contract. Forgetting removes a record's entries from every registered vector epoch, verifies their absence, marks affected epochs dirty, and records the cleanup in a deletion receipt. The sidecar can be discarded and rebuilt from canonical memory. This is more work than maintaining only ordinary lexical indexes, but it prevents semantic reach from silently weakening correction and purge guarantees.

Explanation remains layered. A lexical result identifies matching terms and a graph result shows a typed path. Semantic reports add model and epoch identity, cosine similarity, lexical and semantic ranks, reciprocal-rank-fusion score, and canonical validation. Similarity explains why a record was nominated; it does not explain why the record was trusted or give it permission to affect the world.

The trade remains real. Default lexical recall can miss synonyms, misspellings, multilingual paraphrases, and conceptual equivalence. Optional semantic investigation improves that discovery surface, while the model-facing recall path stays smaller, deterministic, deletable, and inspectable. AETNAMEM claims neither universal lexical superiority nor that an embedding score is evidence of truth.

Important boundary. Embedding retrieval can be reproducible when every required model and index artifact is preserved. Aetnamem's narrower claim is that trusted agent recall does not require those learned-model artifacts; when an investigator opts into embeddings, the additional replay boundary is explicit and the returned record must still pass canonical verification.

8 How AetnaMem Keeps Recall Fast and Governed

Article 8 of 20

FTS5 gives Aetnamem a fast lexical shortlist; lifecycle policy and deterministic ranking still decide what an agent may see.

FTS5 means SQLite Full-Text Search version 5. It is the local catalogue that finds records containing query terms and phrases without asking a model to compare the query with every stored record.

For ordinary agent recall, AETNAMEM asks FTS5 for active lexical matches, fills unused capacity with recent active records, and ranks only that bounded candidate set. The default candidate cap is 200 and the storage layer enforces a hard per-call ceiling of 2,000.

The next step is just as important as the shortlist. Lifecycle eligibility is applied before ranking. Quarantined, superseded, and tombstoned records do not become ordinary recall merely because their words match well. The deterministic ranking combines lexical relevance, source-trust weight, and recency. Those scores are retrieval utilities, not probabilities that a statement is true.

The retrieval event stores a query digest, returned identifiers, algorithm parameters, and a bounded candidate ledger. This keeps routine work and audit evidence bounded together.

FTS5 has a real limitation: “reduce API spending” may not match “lower inference cost.” Version 0.5.1 added optional semantic and hybrid search for human investigation, but the default agent path remains lexical and local.

Important boundary. A candidate cap can exclude a relevant record, and FTS5 can miss synonyms, misspellings, or multilingual paraphrases. Bounded work is an operational guarantee, not universal retrieval quality.

9 Audit Search Without Knowing a Memory ID

Article 9 of 20

An investigator should be able to begin with an ordinary clue and reconstruct the memory story.

Strong evidence is not very useful when only a database specialist can find it. AETNAMEM therefore exposes three investigation commands: `memories` lists what is retained, `search` finds matching memory and evidence, and `trace` follows influence through time.

An auditor can start with “preferred airport,” a session, a date interval, a tool name, a failed outcome, or a memory plane. Search scopes include memories, media observations, source episodes, retrievals, audit events, four-memory runtime runs, and guarded actions. The result can be printed as readable text or exported as structured JSON.

Trace expands matching items through their recorded relationships. A typical story can show admission, quarantine or promotion, retrieval candidacy, returned memory, compiled context, an agent or tool action, and a recorded outcome. If plaintext was purged, the trace keeps identifiers and commitments without pretending it can recover the deleted words.

Investigator searches are read-only by default. An operator can opt into a separate digest-only access chain that records who searched, the query and filter digests, the result-ID digest, count, and semantic epoch. Keeping that chain separate prevents an investigation from altering the agent evidence being investigated.

Important boundary. A subject identifier scopes stored data; it is not authentication. Audit search can expose sensitive retained context and must sit behind a trusted access-control layer in a multi-user deployment.

10 Four Kinds of Memory, Four Different Jobs

Article 10 of 20

Working, semantic, episodic, and procedural memory should cooperate without being collapsed into one undifferentiated text store.

Working memory describes the task now: the goal, progress, constraints, and unfinished state. AETNAMEM stores explicit snapshots keyed by subject, agent, session, and task. Hidden model reasoning is not a supported input.

Semantic memory contains durable governed facts and preferences. It uses the canonical record, trust, lifecycle, provenance, recall, graph, correction, and deletion machinery described earlier in this series.

Episodic memory records what happened: task outcomes and reviewed lessons from past attempts. A failure can create a quarantined lesson proposal, but it does not silently teach every later run.

Procedural memory contains how to perform work. AETNAMEM selects versioned, content-addressed SKILL.md files and records which procedure informed a run. A failed outcome may propose that a procedure needs review; it never rewrites and activates the skill automatically.

These planes have different update rules because they answer different questions. The runtime gathers their contributions behind one interface, applies one context budget, and preserves a manifest showing exactly what was compiled.

Important boundary. Calling four inputs “memory” does not make them correct or useful. Each plane still needs trustworthy source labeling, review policy, and task-specific evaluation.

11 One Context Pack Instead of Four Memory Tools

Article 11 of 20

The agent should receive one bounded context pack, while Aetnamem handles memory-plane orchestration and evidence.

On each turn, the runtime collects contributions from every enabled plane, enforces per-plane and total character budgets, records contribution hashes, and returns one `aetnamem-runtime-pack-v1`. Stable material can be placed in a system prefix while dynamic state stays close to the current turn.

Users do not need to design this configuration from scratch. A ten-step setup wizard and four presets provide understandable starting points: `starter` for one person and agent, `private` for a smaller context, `team` for a trusted multi-agent host, and `benchmark` for controlled evaluation. Presets become ordinary JSON rather than hidden service behavior.

The generic integration remains MCP. Running `aetnamem runtime mcp` keeps the original fifteen memory contracts and adds `memory_prepare_turn` and `memory_record_outcome`. Existing clients can keep using the original catalogue; orchestration is opt-in.

OpenClaw plugin 0.3.1 discovers the orchestration tool and can fall back to its legacy hooks when an older server does not expose it. Grok CLI can connect to the same local MCP runtime. The Python engine, CLI, and MCP remain generic rather than becoming a Grok-only backend.

Important boundary. An outcome submitted through ordinary CLI or MCP is `caller_asserted`. Transporting the value does not prove that a trusted host verified task success.

12 Multimodal Memory Without Storing Media Bytes

Article 12 of 20

Aetnamem governs what an agent observed in media; it does not need to become an image, audio, or video store.

A multimodal host or model can inspect a photograph, recording, video, or document. It sends AETNAMEM a typed text observation plus an evidence envelope: modality, exact-byte SHA-256 digest, secretless host reference, segment, extractor provider, model and version, and optional model digest and confidence.

AETNAMEM stores a first-class artifact row and a quarantined observation record. The observation does not need to match the ordinary fact-extraction grammar. A person or trusted host can inspect and promote it through the same governed lifecycle as other memory.

Confidence stays evidence. A value such as 0.86 from one extractor is not assumed comparable with 0.86 from another, and it cannot promote the record or authorize an action.

The media bytes remain under host control. AETNAMEM does not caption, transcribe, OCR, embed, fetch, retain, or delete them. Semantic investigation can embed the canonical text observation; provenance travels with the result, but no media embedding is created.

Generic CLI and MCP callers are forced to the honest `caller_asserted` digest-assurance tier. `verified_by_aetnamem` is currently rejected because no released path lets the engine hash the original byte stream itself.

Important boundary. A text observation is evidence of what an extractor reported, not proof that the report correctly describes the media. The original artifact and its access controls remain the host's responsibility.

13 Forget the Exact Artifact, and Say Exactly What That Means

Article 13 of 20

Multimodal deletion becomes auditable when artifact identity is an indexed relation rather than metadata buried in JSON.

Every observation links to an artifact identified by subject and the SHA-256 of one exact byte stream. A forget-by-artifact request can therefore find all derived observations without guessing from captions or scanning opaque metadata.

The deletion transaction purges observation and episode payloads, graph derivatives, and registered semantic vectors before tombstoning provenance metadata. It verifies absence before committing and returns an `aetnamem-artifact-deletion-receipt-v1` naming the covered objects.

The receipt is deliberately narrow. It says the digest identifies an exact byte stream, reports whether derived AetnaMem objects were removed, and states that the host file was not deleted. A resized photograph, transcoded video, or re-encoded audio file has a different digest.

Re-extraction has an explicit lineage rule. Artifact plus segment plus extractor identity defines a lineage. A rerun can supersede an older quarantined result from the same lineage, while observations from different extractors accumulate. Promoting a newer result supersedes an older active record from that lineage so recall does not show two accepted versions.

Important boundary. The receipt proves deletion of memories derived from digest X inside the governed store. It does not prove deletion of re-encoded copies, backups, provider logs, exports, or the host-controlled original.

14 The Graph Is a Rebuildable Map

Article 14 of 20

Relationships can recover useful multi-hop memory when every path still leads back to canonical evidence.

Suppose accepted memory says “my boss is Sarah” and “Sarah prefers SEA.” A flight question may match the first statement but not the answer statement. A derived entity-relation graph can follow the boss relation to Sarah and then the airport relation to SEA.

The useful output is not only “SEA.” It is a bounded path showing which entities and typed edges were visited and which canonical records support them. Graph candidates and direct lexical candidates are combined without pretending that their scores have identical meanings.

The graph is a map, not canonical truth. Edges point to records, records point to source episodes, and lifecycle transitions propagate into derived structures. Quarantined edges cannot bridge trusted facts. Corrections supersede affected single-valued relations. Forgetting removes or tombstones the relevant graph objects.

Entity merges remain proposal-only and approved merges use reversible pointers. If extraction logic improves or the graph is damaged, the map can be rebuilt from retained canonical records.

Important boundary. A visible path explains how a result was nominated; it does not prove that every source statement or extracted relation on that path is factually correct.

15 Trace the Influence, Then Verify the Chain

Article 15 of 20

Auditability means reconstructing how memory participated, not just showing that a row exists.

The `trace` command joins retained evidence into a chronological story: memory admission, candidate consideration, retrieval, compiled context, runtime intervention, action, and outcome. It can begin from free text or from a known subject, session, run, record, or event type.

Important transitions also append to a per-subject SHA-256 hash chain. Each event commits to the previous event hash, so an edit, insertion, or interior deletion breaks local verification. Semantic investigator access can use its own digest-only chain, preserving separation from the behavior being audited.

A local chain has an important limitation. An attacker controlling the whole database could truncate the tail or replace the chain and recompute it. AETNAMEM therefore supports external checkpoints that move the event count, sequence, and head hash into another trust domain.

Trace and chain verification answer different questions. Trace explains the recorded relationships. The chain checks local continuity relative to a trusted starting point or checkpoint. Neither proves that the original input was truthful or that an external action really happened.

Important boundary. Deleted plaintext stays deleted. An audit may retain identifiers, digests, lifecycle events, and receipts while correctly refusing to recreate purged content.

16 Information Is Not Authority

Article 16 of 20

A remembered instruction can inform a plan, but only trusted authority can approve an external effect.

An assistant may remember that a report belongs in a particular folder. That memory can help prepare a proposal. It should not authorize the assistant to overwrite the file.

AETNAMEM separates recalled evidence from external authority. Guarded actions prepare a canonical plan containing the operation, arguments, preview, preconditions, adapter manifest, dependencies, evidence references, and policy version. Approval binds to the digest of that exact plan for a limited period.

Before execution, the engine rechecks the plan and approval commitments, adapter identity, policy, and resource preconditions. The reference filesystem adapter stops if a file changed after review and reads the result back after a write to verify the expected digest.

This boundary also limits durable prompt injection. Quarantined web or tool content may be retained for review, but it cannot manufacture the user authority required to execute a protected operation.

Important boundary. This protection applies only when model-requested effects are routed through the registered broker. Direct shell, filesystem, database, or provider access can bypass it and must be restricted by the host.

17 “Uncertain” Is a Responsible Result

Article 17 of 20

When an external call may have succeeded, pretending failure and retrying can be more dangerous than admitting uncertainty.

Database operations can often commit or roll back atomically. External calls cannot. A provider may receive a request, perform the effect, and disconnect before returning a response. The caller sees an error but does not know the outside state.

AETNAMEM persists execution state before dispatch and does not hold a SQLite transaction open across the external call. If the call raises after dispatch, the action becomes uncertain. If the process restarts during a committing or compensating operation, the transaction becomes recovery required.

Provider-specific idempotency lookup or authoritative state inspection may later resolve the outcome. Generic code does not retry blindly, because a second request could duplicate the first effect.

For multi-step plans, previously verified effects may be compensated in reverse order. Compensation runs only when current state still matches the engine’s verified after-state. If another actor changed the resource, the engine refuses to overwrite that later work.

The final result may be committed, compensated, partial, uncertain, or recovery required. These states are less tidy than success and failure, but they preserve what the system actually knows.

Important boundary. A generic control plane cannot resolve ambiguity when a provider offers neither idempotency records nor an authoritative way to inspect the resulting state.

18 Memory Impact: Did Remembering Actually Help?

Article 18 of 20

Retrieval is not impact; causal evidence requires controlled interventions and outcomes verified outside the agent.

If an agent succeeds after receiving a memory, it may have succeeded without it. Relevance scores, usage counts, and correlations cannot answer the cause-and-effect question by themselves.

Version 0.6.0 introduces the default-off Memory Impact Lab. Its technical foundation is the Causal Memory Ledger: a record of candidate contribution hashes, assignment probabilities, assigned and applied memory arms, experimental stratum, policy identity, seed commitment, and final context manifest.

The registered design tests all sixteen combinations of working, semantic, episodic, and procedural memory. Within each task block, every arm from 0000 through 1111 appears once in randomized order. Exact exposure evidence fails closed if an assigned contribution is missing or truncated.

Grok does not grade Grok. The controller clones the frozen SQLite snapshot and workspace, compiles context on the host, runs the agent in a restricted sandbox, and invokes a hidden verifier outside the agent workspace. A signed receipt binds the assignment, context, output, workspace, verifier digest, cost and outcome metrics.

Before any paid model study, the synthetic stage plants known plane effects and deliberate confounding. It checks estimator error, direction, interval coverage, null false positives, and whether randomized conclusions outperform naive observational attribution.

Important boundary. The laboratory can measure a registered intervention; its existence does not prove that memory improved an agent. Adaptive production selection remains off until a frozen policy wins on held-out tasks and replicates.

19 What the 100-Run Grok Pilot Actually Showed

Article 19 of 20

A promising pilot is useful when its evidence and incompleteness are reported together.

The corrected paid Grok 4.5 recovery pilot deliberately stopped after 100 registered calls. It produced 100 signed receipts, 57 host-verified successes, zero aborts or timeouts, and six complete balanced sixteen-arm blocks.

Across those six complete blocks, the all-four-memory arm succeeded in six of six tasks and the no-memory arm in one of six. Procedural memory had an exploratory success-effect estimate of +0.1875, with a 95 percent interval from +0.049631 to +0.325369. The experiment recorded USD 8.3339928 in provider cost, plus USD 0.016532 for the corrected smoke test.

The evidence pipeline did what it was designed to do. Schedule integrity and the seed reveal verified. There were no failed, aborted, or extra receipts. Protocol, schedule, Grok executable, metrology, public key, verification JSON, and report digests are published in the repository's compact evidence summary.

The scientific result is intentionally narrower. The registered schedule contained 384 calls, so 284 remained missing after the cap. Only six balanced blocks were complete. The frozen-policy held-out phase did not run. The wide intervals for most plane effects reflect that limited evidence.

The defensible conclusion is therefore not "AetnaMem makes every Grok task better." It is that AETNAMEM successfully controlled which memory Grok saw, measured externally verified outcomes, and preserved evidence for a larger confirmatory study.

Important boundary. The 6/6 versus 1/6 arm result is exploratory, not a deployment recommendation or general causal product claim. Completion, held-out benefit, and independent replication are still required.

20 The End Goal: An Evidence-Based Memory Control Plane

Article 20 of 20

Aetnamem should spend context where memory earns its place and keep the evidence needed to audit that decision.

AETNAMEM is designed so that memory does not belong to one language model. Grok, OpenClaw, Hermes, or another MCP client can use the same local engine. The model may change while canonical records, observation provenance, runtime manifests, interventions, outcomes, deletion receipts, and audit evidence stay outside it.

The current product combines several roles:

- a governed semantic record store with quarantine, correction, graph lineage, and verifiable deletion;
- a four-memory runtime that compiles working, semantic, episodic, and procedural contributions under one budget;
- a multimodal evidence layer that stores observations and artifact provenance without storing media bytes;
- an investigator surface for lexical, semantic, hybrid, and chronological trace search;
- a beta Safe Switch path that captures beside an existing OpenClaw or Hermes agent, requires human review, previews exact context, limits the first exposure, and can restore the saved host configuration;
- an effect-control boundary that keeps remembered information separate from external authority; and
- an experimental laboratory that asks whether offering memory changed a verified outcome enough to justify its context and cost.

The strongest positioning is therefore not merely “four kinds of memory.” It is: **AetnaMem is an evidence-based memory control plane. It measures which memory policies improve an agent, preserves the proof, and aims to spend context only where memory earns its place.**

That last sentence is a direction, not a completed universal claim. Version 0.6.1 makes reversible adoption the beta product: observe, review, preview, canary, then activate or roll back. The Memory Impact laboratory remains research. Its measurement spine and exploratory Grok pilot do not yet prove general benefit. The next scientific gates are a complete powered schedule, a frozen held-out policy win under the same budget, and replication.

Important boundary. AETNAMEM is evolving open-source infrastructure. Judge it by released behavior, signed and reproducible evidence, explicit failure states, and claims that become stronger only when the corresponding gates pass.